

Referee comments PLB

<https://arxiv.org/pdf/2606.02067>

Reviewer #1:

Dear authors,

Thank you for this nicely well-written paper. The manuscript is well-written, the results are interesting and I believe the paper deserves to be published after some minor revisions. Please find my detailed comments below.

Introduction:

In the SM, the production of a $t\nu$ system with an associated n -quark is SM suppressed. ->I feel the statement is too generic. There are SM processes (e.g. top quark decay) where there is no issue. So it would be good to be a more precise here. I assume it is referring to b -meson decays.

→ We meant the process $gq \rightarrow Wb \rightarrow t\nu b$, rather than all SM processes yielding a $t\nu + b$ final state. This process is CKM-suppressed by the small V_{ub} and V_{cb} . We have rewrited it as:

“In the SM, the production amplitudes of a $t\nu$ system with an associated b -quark, which can directly interfere with the BSM signal, are CKM-suppressed. Consequently, the SM-BSM interference is expected to be small in the high-mass region.”

Section 3

Signal model footnote. This is a bit confusing to me. You say that the β_{13} is roughly $0.2\beta_{23}$ and then you say it is ignored. Does this mean that you actually put it to 0 in the end?

-> No, β_{13} in our signals is computed as $V_{td}/V_{ts} \cdot \beta_{23}$. It is omitted from Eq. (1) since it has a negligible impact of $|(V_{cd} \cdot \beta_{13}) / (V_{cs} \cdot \beta_{23})| \sim 0.05$ on coefficient CLL.

Section 5

Why is there no muon veto for hadronic tau candidates?

→ the probability of a muon to be mis-identified as a had tau candidate is much smaller than that for an electron. Therefore we only apply eVeto RNN for tau candidates.

Define a calorimeter-tagged muon before

→ Will define a calorimeter-tagged muon:

“calorimeter-tagged muons (identified from ID tracks matched with calorimeter energy deposit compatible with a minimum ionizing particle) sharing an ID track with an electron”

Why is for an electron only the closest jet vetoes, while for muons all jets close by are vetoed if they have max. 2 tracks?

→ Not all jets near a muon are vetoed. We only reject a jet to which a muon is ghost-associated to within $\Delta R=0.4$ and it has not many associated tracks. Electrons typically produce sizable calorimeter showers, while muons are usually minimum-ionising. This leads to the different strategy for jet-electron and jet-muon overlap removal. We agree that the current wording is a bit misleading and propose to revise it to:

“jets within $\Delta R=0.4$ of a muon if they have at most two associated tracks and the muon is ghost-associated with the jet.”

ETmiss is redefined here while it was defined before

Will remove “The magnitude of this vector is denoted by ETmiss.”

Event selection

In general it could use some more motivation for some of the cuts. Now they are mostly listed. It is nice to connect the requirements to the background which is being rejected (e.g. why max. 4 jets). What kind of criterion was used? I see that the stat. unc. on bkg is kept small but assume there was also a specific criterion used to extract signal?

→ The jet multiplicity is required to suppress top quark background. The high-jet pT and the invariant mass requirements are used to select events consistent with an on-shell LQ decay, whereas the large mT, mET and $\Delta\phi$ requirements target the high-mass, back-to-back taunu topology for the non-resonant LQ process.

The specific value for each selection were determined through the SR optimisation by scanning the selection thresholds and maximising the expected significance, while requiring the statistical uncertainty on bkg. remain below 30%.

For the m_{jthad} the light jet is a proxy for the b-quark, otherwise it could not give the leptoquark mass. What is the purity of this selection? Have you checked that it would not be better to use the highest GN-2 value instead? Maybe that would have more purity?

→ The light jet in SR0b is not selected simply as a proxy for b-quark. For large β_{23} , the SR0b-Res has a sizeable contribution from resonant $U1 \rightarrow \text{stau}$ decay. In particular, $U1 \rightarrow \text{charged lepton}$ decay width for $b\tau:\text{stau}:\text{dtau}$ is approximately $1:|\beta_{23}|^2:|\beta_{13}|^2$, which means that fraction of stau decay increase from ~3.8% to ~74% when β_{23} increase from 0.2 to 1.8. Selecting the light jets with highest GN-2 value is therefore not appropriate for the SR0b.

Background estimation

Why are we only considering fully leptonic $t\bar{t}$ and tW decays? Explain why this is dominant. Also, we estimate both together, but they must behave quite a bit different in

different bins and also in the extrapolation? How is that dealt with? Do you have a ratio uncertainty for them?

→ We studied the inclusive $t\bar{t}$ and single top production and confirmed that the leptonic $t\bar{t}$ and tW decays constitute the majority of top background since we require large missing ET. The single-top/ $t\bar{t}$ ratio was also compared between the SRs and CRs and found to be consistent within the MC statistical uncertainties.

I would swap $t\bar{t}$ and W-jets section to be consistent with the discussion before.

For the W+jets section, in the final hadronic tau case, there will be extra neutrino's wrt leptonic case? How do you make sure this does not mess up your modeling?

→ We have confirmed the $W \rightarrow e/\mu \nu$ to $W \rightarrow \tau \nu$ extrapolation in the low-mass $W \rightarrow \tau \nu$ region. The data and simulation show good consistency.

Systematic uncertainties

30% uncertainty on W+HF vs. W+LF. Was a cross-check made for this analysis too? Why would these old ATLAS studies be valid for this analysis, in another phase space. It would be good to justify this.

→ As shown in Refs. 87-90, mismodeling of W+HF/LF ratio depends on the event kinematics indeed. The 30% uncertainty is chosen conservatively based on the largest deviations observed in the previous ATLAS studies.

Since the analysis is stat-dominant, further increasing this uncertainty would have a negligible impact on the final result. We also decorrelated this uncertainty between the resonant and non-resonant regions to account for possible kinematic dependence.

For the $Z \rightarrow \nu\nu$ background, why was the choice made not to use a fake tau method, like almost all similar searches do? We know that the modeling is bad and it is confirmed by the 60% differences you are already seeing. Would it not be better to correct using such a method? Now we apply a 100% uncertainty on a MC method. The 100% is also not really motivated.

→ The contribution from the fake background in this analysis is negligibly small. Even the 100% uncertainty would have less than 1% impact on the final result. So we decided not to use a dedicated fake tau method for this background. The 100% uncertainty is assigned conservatively from the fake study.

Results

Motivate why you completely uncorrelated the W normalization (same for top) in the two regions. Connected to this. For top the two factors are roughly similar, but for the W they are very different. Do you understand why the W normalization in both regions would be so different?

→ We decorrelated the normalization in the Res/Non-Res region since they probe different kinematics regimes. The difference between W+jets normalisation factors can be explained as a result of the different jet- p_T range in the two regions.

It is mentioned that it is verified that the total yield stays positive in all cases, but is this also the case during the limit setting when μ is allowed to vary or could weird things happen there? Do you constrain how much μ can vary?

→ The contribution from BSM+interference could become negative in case the μ is small enough. But in the parameter region where the analysis is sensitive to, it does not affect the upper-limit determination since the interference contribution is small compared with the pure BSM contribution.

Discrepancy in Validation Region. For this discrepancy, could it be related to the big difference in normalization factors? If there is a specific distribution that convinced you that it is a statistical fluctuation, consider adding it to the supplementary material

→ The mET distribution and dedicated check in the low mass VR convinced us that it's a statistical fluctuation.

I am also missing a bit of information about the correlation model of the nuisances. Which ones are correlated between signal and background, across bins, across years?

→ the nuisances parameters are generally correlated across regions and between signal and backgrounds. Theoretical uncertainties are processed specifically, and W+jets HF modeling uncertainties are decorrelated between resonant and non-resonant regions. The fit is performed with one bin per region, and no special treatment is applied across the years.

Reviewer #2:

First of all, congratulations on this interesting result by ATLAS. The overall quality of the document is really good. I have some comments that will help you improve your paper. Please find them below:

page 2 last paragraph: Two types...

In the resonant and non-resonant categories, do you consider only the signal that each category is dedicated to?

Similarly, for the b-tagging categories. What is considered as signal in each category and what do you do for the other signals? Is each signal extracted from the dedicated category?

→ We do not separate signals according to the categories. The resonant and non-resonant SRs are optimised for the corresponding topologies, while both can contain contributions from the other topology (although usually much smaller). The same applies to the b-tagging/b-veto categories.

page 4 4th paragraph:

How reasonable is it to look only in the left handed scenario?

→ The pure left-handed scenario already probes the $b \rightarrow c\tau\nu$ transition and is therefore chosen as the main benchmark scenario. A benchmark with a sizable right-handed coupling, $\beta_{R33}=1$, has been also studied. We found that it enhances the non-resonant process more than resonant process and lead to a better sensitivity at high LQ masses. The upper limit on the combined resonant and non-resonant visible cross section therefore show

the dependence on the LQ mass. We could add a short description of this study in the text, something like "The impact of a sizable right-handed coupling is also investigated and is found to enhance the non-resonant process relative to the resonant one, resulting in stronger sensitivity at high LQ masses."

page 4 last paragraph/ page 5 first: The relative contributions of...

Can you explain more the interference between resonant and non resonant signal and how this is taken into account in the signal extraction.

→ All resonant and non-resonant signals are included simultaneously in the matrix-element calculation. Therefore, any interference between resonant and non-resonant diagrams is naturally included in the nominal BSM samples, and no specific treatment is needed in the signal extraction.

page 5 second paragraph: no hadronic tau triggers? a bit surprising, you may want to comment.

→ The missing ET trigger is already fully efficient for the signal-region selection, since all SRs require missing ET > 200 GeV. Therefore, no dedicated hadronic tau trigger is needed for the signal efficiency. Single lepton triggers are used for CRs and VRs.

page 5 third paragraph: The contribution from interference between the signal process and the SM ...

Can you explaining more this interference as well? Later on (page 11 in the result) you mention that the μ is set to $\sqrt{\mu}$, it is not obvious why.

→ The interference occurs between the BSM and SM tauon amplitudes.

Two kinds of samples are used, corresponding to the pure BSM term (S) and the interference term (I), which scale with μ and $\sqrt{\mu}$ in the fit, respectively. And the signal model is therefore treated as $(\mu * S + \sqrt{\mu} * I + B)$ in the likelihood.

page 6 and page 9 second paragraph:

The names of the working points don't mean much to those of us out of ATLAS. This is especially true on page 9. Either numerical description of the efficiency or simply the mention that a "tight" selection is applied to ID and Iso is enough. My understanding is that the preselection leptons are used for the overlap removal, while the CR is tighter. It seems to me that this is fairly standard practice.

If there is another point that I have missed, and requires the exact names? If so make it more clear because as an educated reader I didn't get it.

→ The understanding is correct.

page 11, results, first paragraph:

The interference is not very well described, appears almost out of context here. It is implied earlier but not enough information is given. also this is interference between the signal and the SM bkg right?

Is there also interference between resonant and non-resonant signal?

Why $\sqrt{\mu}$?

→ Already addressed for the similar question.

page 11, results, third paragraph:

can you quantify satisfactory?

→ The post-fit background predictions agree with data within approximately 1sigma in all VRs except VRW-NonRes, where a deviation of about 2sigma is observed.